SEAT NO. CT-23025

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(Computer Science)

SPRING SEMESTER EXAMINATIONS 2026

Time : 3 Hours | **Batch 2023** | **Dated : 08-JUN-26**
Max Marks : 60

Theory of Programming Languages - CT-367

Note: Attempt all questions. All questions carry equal marks.

Q.1 (a) **Categorize** and **describe** the primary programming paradigms. **Perform** a comparative analysis between Imperative and Declarative languages. Illustrate your answer with a representative code snippet for each. (CLO-1) **5**

(b) **Examine** the role of Syntax Analysis within the front-end of a compiler. **Demonstrate** the application of Context-Free Grammars (CFG) in the derivation of a parse tree for a specific input string, using a step-by-step example. (CLO-2) **5**

Q.2 (a) **Identify** and **explain** the critical design issues associated with arrays. Provide a structured classification of array categories based on their operational characteristics. (CLO-1) **5**

(b) **Analyze** the multi-phase process of source code compilation. Provide a detailed, labeled schematic diagram illustrating the flow from Lexical Analysis to Code Generation. (CLO-2) **5**

Q.3 (a) **Classify** variables based on their lifetime (binding to storage). **Contrast** the advantages and disadvantages of each category regarding memory management and execution efficiency.(CLO-1) **5**

(b) **Evaluate** how grammar ambiguity impacts the reliability of the compilation process. Using a mathematical expression as an example, **demonstrate** how multiple parse trees can lead to different results, and propose a solution using precedence or associativity rules. (CLO-2) **5**

Q.4 (a) **Explain** the mechanism of short-circuit evaluation in logical expressions. **Identify** modern programming languages that implement this feature and describe how it impacts program performance and safety. (CLO-1) **5**

(b) **Investigate** the relationship between Referential Transparency and Functional Side Effects. **Explain** how the presence of side effects can violate referential transparency in a program, supporting your discussion with comparative examples. (CLO-2) **5**

Q.5 (a) **Discuss** the significance of Operand Evaluation Order in complex expressions. **Explain** the rules governing these evaluations and analyze how Type Conversion (Coercion vs. Casting) affects the final evaluated result. (CLO-1) **5**

(b) **Construct** a lexical analyzer using Flex (**.lex**) and a syntax analyzer using Bison (.y) to process arithmetic expressions involving identifiers, constants, and operators (+, /). The implementation must successfully validate expressions like (**ID + CONST**). (CLO-2) **5**

Q.6 (a) **Contrast** the concepts of Pointer Value versus Pointer Address. Explain how the lifetime of a pointer variable itself differs from the lifetime of the heap-dynamic variable it references.(CLO-1) **5**

(b) **Explain** the "Dangling Pointer" problem that occurs when a function returns a pointer to a local (stack-allocated) variable. **Discuss** the possible solutions and their pros and cons. (CLO-2) **5**

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